Make the kitchen easier. Prove it in one meal.

Product, marketing and engineering review | 12 October 2026 | Prepared for Matt Aldous

Know what you have. Decide what to cook. Buy only what you need - for everyone at home.

Recommendation: keep the broad vision, but make one complete household routine excellent before promoting breadth. OffMenu already contains much of the proposed feature surface. Its present constraint is trustworthy, usable data and continuity between screens, people and devices.

Evidence from this review	What it means
40 recipes; only 1 fully quantified	The only fully quantified recipe is the authored demonstration. The 39 imported recipes all contain unresolved ingredient identities. Prioritise a small, validated catalogue.
47 existing tests passed	41 JavaScript tests and 6 Python tests pass. This is useful code evidence, not proof of camera reliability, household sync or clinical accuracy.
Manual cloud snapshots	Users must save explicitly. Family profiles are records inside one account; they are not independent household memberships.
Consumption and inventory are separate	After-meal records do not deduct ingredients. Saved leftovers do not yet become usable cooked stock. The main promise therefore remains incomplete.

The proof of concept to promote

A person scans or adds five useful foods, sees three realistic dinner options, picks who is eating, adds only the missing ingredients to a shared list, cooks, and optionally records leftovers or completion. Their second device shows the same result. The whole kitchen never needs to be catalogued first.

Three decisions

- **Lead with dinner and household relief.** Keep the playful home-checkout identity as the memorable interaction, while letting users start from meals, shopping or a single shelf.
- **Ship licensed, feature-ready data.** Start with 30 validated meals and roughly 100 useful ingredient mappings; expand in measured regional packs. These are proposed initial targets, not current inventory.
- **Make health and automation optional layers.** Add them through clear consent, provenance and scoped access after the everyday routine works.

Scope: current source archive version 4 and the live public landing page/demo were reviewed. No account sign-in, physical camera, paid provider, wearable or third-party OAuth client was certified. Proposed milestones and commercial ideas are hypotheses, not measured demand or delivery commitments. [1, 2]

What the current product really does

Implemented capabilities are distinguished from live verification and proposed improvements.

Area	Observed implementation	Priority gap
Capture and shelves	Continuous ZXing camera scan, manual barcode, staged receipt, location-aware rebuild/unpack/cleanup, undo of staged lines.	Camera defaults to front-facing. Add a visible camera switch, no-barcode foods, and test physical devices.
Meals and shopping	40 recipes, quantities, household filters, weekly requirements, stock-aware gaps and export formats.	334 of 375 ingredient entries lack quantities. All 39 imports have unresolved identities; customer-facing lists contain review work.
Dietary matching	Small keyword/regular-expression rules; avoids and diet preferences.	Reproduced: turkey is accepted for vegetarian; peanut butter is rejected as dairy. Replace this with evidence-aware ingredient rules.
Nutrition	12 nutrients, available-product estimates, partial coverage, frozen meal logs, completion percentage.	Uses first matching stocked product; lacks generic composition fallback and allocation to actual cooked lots.
Household and sync	Optional member metrics; one per-account JSON snapshot, revisions and conflict rejection.	No independent memberships, auto-sync or concurrent merge. Current whole-snapshot writes have a 2 MB request limit.
Consumption and waste	After-meal logs and partial cleanup removal. Estimated check reminders are separate from entered expiry.	Meal logs do not change stock. Leftover status is contextual, not a cooked-food lot. Batch/portion linkage is needed.
Interoperability	REST reads, remote MCP reads, personal tokens, OAuth/PKCE, revocation, exports.	Reads return saved cloud state. Real-client interoperability, household-scoped permissions and freshness UX remain unproven.
Presentation	Warm green/coral identity, no-account demo, large controls, reduced-motion rules, service-worker shell.	Eight workspace navigation actions and developer/provider details compete with the main tasks. Imperial selection does not convert quantities.

Verification and immediate interpretation

The live demo displays 40 menus and one ready dish with its supplied sample stock. Source inspection explains why many cards ask the visitor to review ingredient quantities. Existing tests pass, but the two dietary counterexamples above also reproduce: coverage is incomplete, so green tests must not become a safety claim.

Additional planning issue: stock is filtered against today before weekly summaries, rather than allocated against each future cooking date. Add date-aware stock allocation before promising that a future meal can be made from current supplies. [2]

A market position worth testing

The reviewed competitors already cover substantial parts of the concept. This is a selective current comparison, not a claim that none has an overlapping feature.

Product	Established overlap	Implication for OffMenu
Samsung Food / Food+	Recipes, meal plans, shopping, nutrition and food-list/expiry-aware discovery. [3]	“All in one” and “AI meal planning” will not distinguish the product.
Paprika	Recipe management, pantry, shopping and device sync; Paprika 4 open beta advertises barcode scanning. [4]	A scanner plus recipe manager is already a competitive baseline.
AnyList	Shared groceries, meal calendar, recipes and instant household updates. [5]	Household collaboration must feel immediate and ordinary.
Cronometer	Barcode food logging, nutrition, exercise and health information. [6]	Do not compete first on nutrient count or medical precision.
SuperCook	Recipes using available ingredients and missing-ingredient discovery. [7]	Ingredient unlocking alone is useful, but not a defensible novelty claim.

The defensible product hypothesis

A household memory of food: what arrived, where it is, who can use it, what was cooked, what was actually eaten, what remains and what should be bought next. The value is lower effort and better decisions across that loop. This is a positioning hypothesis; research here does not establish patent novelty or exclusive differentiation.

First customer segment	Job to solve	Best demonstration
Busy, mixed-preference households	One dinner with practical adaptations and fewer duplicate purchases.	Choose the people present; get one base dish plus separate toppings or sides.
Budget- and waste-conscious cooks	Use what is already owned before it is forgotten.	A use-first dinner and a short, explainable shopping gap.
Organisers and technical early adopters	A portable kitchen record that works with existing tools.	Open exports and an authorised assistant answering a real inventory question.

Commercial boundaries

Start with Australian everyday households and a clear supported-device list. Expand by evidence of repeated use. High-risk allergies, clinical meal prescriptions and care-provider workflows need stronger validation and governance before becoming lead acquisition claims. Retain engineering power in a separate developer area.

One easy routine, five entry points

Proposed information architecture: Today | Kitchen | Plan | Shop | Me. Keep a persistent Scan/Add action; place documents and developers under Me.

Moment	Customer experience	What the system does
First visit	Try a stocked demo, scan/add a few foods, or choose dinner. Ask only who is eating and essential restrictions.	Keep guest progress; request an account when saving or inviting has an obvious benefit. Make guest-to-account transfer explicit and reversible.
Add food	Scan barcode, search produce, repeat a recent item, type or speak, or review a receipt photo.	Stage results immediately; enrich in the background. Keep unknown items usable with a private name, unit and quantity.
Decide dinner	Show three strong options: ready now, use first, or buy one extra item. Show time, servings, equipment and reasons.	Apply restrictions before ranking. Separate unknown suitability from known conflict. Do not bury the decision under forty incomplete cards.
Shop together	One shared list, aisle grouping, quantities, preferred shops, substitutes and collected state.	Subtract allocated stock once. Show price date, shop and pack size. Distinguish a shopping intention from received inventory.
Cook	Readable steps, timers, keep-screen-awake where supported, scale servings and confirm substitutions.	Create a cooking batch and deduct its ingredient lots once. Record actual ingredient changes for nutrition.
After eating	Optional quick choices: all / most / half / little / none, with a precise percentage if useful. Save, share or discard the remainder.	Split portions between diners and remaining cooked stock. Permit skip; unknown is not zero. Never reward a child for clearing a plate.
Cleanup	Choose a location, scan or select food, enter whole/part quantity, review and undo.	Record discard reason, reconcile stock, and learn a review reminder. Do not infer safety or silently discard food.

Progressive depth

Default cards answer “What can I do next?” Expand for nutrition, source data, uncertainty and detailed history. Offer a quiet mode, large-type mode and simple task view. Remember preferred location and input mode; keep Back, Cancel, Undo and offline status consistent.

Revise the first impression

Keep the existing colours and cheerful register illustration. Replace the shelf-clearing prerequisite with an immediate dinner outcome. Use realistic meal imagery with appropriate rights; keep product photography and AI illustration clearly distinguishable. Label sample data throughout the demo.

Priority feature portfolio

P0 = required for a persuasive public proof of concept. P1 = next retention release. Features can be rich internally while the default interface stays small.

Priority / feature	Customer value	Acceptance evidence
P0 - Validated meal pack	A useful dinner appears immediately.	30 fully quantified, reviewed meals; 100 useful ingredient mappings; source and change history; no unresolved hero-demo recipes.
P0 - Produce and missing products	Everything edible can be tracked, including leftovers and market purchases.	Non-barcode IDs; rapid quantity presets; label photo/manual correction; unknown scan never blocks the queue.
P0 - Ingredient and unit engine	Packaged food matches the culinary ingredient correctly.	Aliases, preparation state, portion basis and dimensional conversions; uncertain density/count conversions stay explicit.
P0 - Household membership	Different people use their own phones and logins.	Owner/editor/viewer roles; invitation and removal; separate profile ownership; no cross-household access.
P0 - Automatic sync	Updates survive device switches and shopping together.	Durable local queue, retry-safe operations, visible sync state, deterministic conflicts and two-device proof.
P0 - Cooking and leftovers	Inventory remains credible after a meal.	One cooking batch deducts once; leftovers become lots; serving, consumption and waste reconcile without double counting.
P0 - Restriction certainty	Users understand known conflicts and missing evidence.	Separate allergy, intolerance, religion and dislikes; blocked / needs check / no known conflict states; package-level evidence.
P0 - Usability and trust	A household can succeed without training.	Keyboard and screen-reader journey; real Android/iPhone camera trial; privacy, deletion, attribution and recovery controls.
P1 - Receipt and label capture	Less manual bookkeeping after shopping.	User-reviewed OCR, duplicate receipt detection, pack/price/date extraction, minimal image retention and no auto-confirmed purchases.
P1 - Better planning	Dinner fits attendance, time, equipment, leftovers and budget.	Reusable weeks, batch cooking, lunchboxes, meal swaps, date-aware stock reservations and optional staple defaults.
P1 - Useful feedback	The app remembers household preferences.	Explicit likes, texture and spice preferences, optional completion history, repeated corrections and easy reset.
P1 - Shop and share	Other people can act on the plan.	Aisles, family checklist, substitutions, list links, printable views, calendar export and regional retailer handoff.

Prioritise measured time saved and repeated completed meal journeys. A new integration should earn its place by improving those outcomes, filling a measured data gap, or enabling a committed customer.

Expansion opportunities with real value

P2 = add after repeat use is established. Discovery = validate access, rights, economics or clinical requirements before promising it.

Opportunity	Useful scope	Timing / constraint
Taste, culture and household rhythm	Cuisine, language, religious preferences, spice/texture, who is home, fasting windows chosen by the user, multiple homes and caregiver handover.	P1/P2. Distinguish preference from certified halal/kosher status. Never infer religion or medical needs.
Inventory confidence	Opened/frozen/thawed states, lot dates, quick “still here?” checks, partial packs, bulk food, decanted containers and private QR labels.	P1/P2. Keep uncertainty visible; avoid requiring gram-level tracking for every household.
Recipe ownership	Private recipes, URL bookmarks/import where permitted, family collections, meal templates, photo notes and creator packs.	P1/P2. Private import is not automatic permission for public redistribution.
Budget optimisation	Unit price, pack rounding, leftovers from pack sizes, basket alternatives, fees, preferred store and maximum extra stops.	P2. Historic observations cannot be advertised as live cheapest checkout or guaranteed savings.
Waste reduction	Use-first queue, quantity/value discarded, reasons, repeated overbuying, suggested smaller pack and seasonal meal rotations.	P1/P2. Saved-food and cost claims need a clear measurement method; avoid unsupported environmental precision.
Nutrition and hydration	User goals, nutrient coverage, drinks, snacks, restaurant/manual entries, recipe substitutions, portion and trend views.	P1/P2. Keep missing nutrients and unlogged days unknown. Show individual and household totals separately.
Exercise and body metrics	Optional activity, steps, exercise schedule, weight/height trends and manually entered measurements.	P2. Date, unit, source and consent per measurement. Do not auto-prescribe energy deficits from a wearable.
Wearables and scales	Health Connect, HealthKit, smart scales, selected vendor APIs and import files.	Discovery/P2. Native permission bridge or approved vendor access; deduplicate overlapping device data.
Safety and recalls	Country recall notices, batch/date matching, clear source links and user review.	P2. Potential match alerts only when identification is incomplete; never imply exhaustive recall coverage.
Assistants and home tools	Read kitchen via MCP; approved list/plan changes; calendar, shortcuts, voice, Home Assistant and household displays.	P1/P2. Existing reads first. Separate household/profile scope, proposed action preview, confirmation and audit for writes.
Care and professional sharing	Time-limited dietary reports, consented dietitian access, accessible carers’ views, food-texture plans entered by professionals.	Discovery. Separate clinical/IDDSI, disease management, supplements and medication workflows from general wellness.
Community and partnerships	Creator meal packs, cooking challenges, opt-in recipe sharing, brand data corrections, retailer referrals and workplace pilots.	P2. Moderate shared content, clear rights and sponsorship, no children’s health leaderboards or health-data ad targeting.

Defer automated grocery checkout, disease treatment, glucose/medication dosing, smart-fridge hardware dependence and a full social network. They add disproportionate obligations before the main household value is proved.

Expand the data that improves decisions

Separate product identification, ingredient identity, composition, recipes, offers and safety information. No single database supplies all six reliably.

Source / opportunity	Recommended use	Rights and operating decision
Open Food Facts [8]	Primary packaged-product lookup; country-specific working set; taxonomy and correction feedback.	ODbL database / DbCL contents; images have separate rights. Preserve provenance and required attribution/share-alike. Measure Australian scan hit rate.
FSANZ AFCD [10, 11]	Primary Australian generic-food composition: produce, raw ingredients and unbranded foods.	Use the database-specific licence, not the website footer. Preserve limitations, Australian-data notice and derivative obligations. Stage a useful subset first.
USDA FoodData Central [12]	Generic fallback and US-market expansion; nutrient and food identity crosswalks.	Public domain/CC0. API requires a key; bulk downloads are also available. Preserve dataset/type/version and market differences.
Authored recipes + Wikibooks [13]	A small reliable starter catalogue, then attributed cultural variety.	Current imported dataset declares CC BY-SA 4.0. Preserve original recipe identity, authorship, changes and licence; verify image rights separately.
GS1 NPC / Trusted Content [17]	Supplier-maintained product identity and richer Australian label content.	Commercial/permission-based access inquiry. Confirm consumer-app display, redistribution, retention and supplier publication rights.
Open Prices [9]	Dated price observations and personal receipt comparison.	ODbL; location/time provenance essential. Coverage is uneven and not a stock or checkout-price service.
CoFID / Ciqual / NZ FOODfiles [18-20]	Future regional composition packs selected by actual users.	Different national definitions and terms; do not treat every “free download” as the same licence. NZ terms need a complete access review.
FatSecret / Edamam / Spoonacular [14-16]	Optional coverage benchmark or contracted service where it improves a measured gap.	FatSecret free editions are US-only in the reviewed table. Edamam restricts caching; Spoonacular restricts transformed storage and competing use. No mirror by default.
FSANZ recalls / FoodKeeper [30, 31]	Recall references; attributed storage guidance and check reminders.	Verify feed, reuse and geographic applicability. Storage guidance never overrides the package or actual handling conditions.
User and manufacturer input	Corrected labels, private recipes, portions, receipts, preferences and product changes.	Private by default; explicit contribution consent for publishing. Verify contributor rights before sharing images or recipes.

RecipeNLG and generic scraped recipe dumps are not approved here for promotional/commercial use: a dataset title or research licence is not sufficient clearance. Large recipe volume is a poor success metric when quantities and rights remain unresolved.

Integrations: connect only what is useful

Access status is as researched on 12 October 2026. An API page or outbound link is not an active integration.

Integration	Customer benefit / next step	Launch treatment
Australian retailers: Coles, Woolworths, ALDI, IGA; WhichGrocer	Test licensed product/offer coverage for the household shopping list. Obtain terms covering store identity, GTINs, timestamps, retention and display.	Outbound links and manual receipts work now. No general authorised consumer stock/cart feed was verified. WhichGrocer terms could not be fully read in this review. [34]
DoorDash [21]	Delivery handoff where supported; consider a partnership if demand warrants it.	Marketplace APIs are limited-access merchant/partner integrations, not a verified public consumer barcode-price catalogue.
Instacart [22]	Shoppable recipes/list pages and retailer discovery in supported markets.	Developer key and supported region required. Prove the precise flow and commercial terms before advertising connected checkout.
Google Health [23]	Potential consented activity/body-data connection.	Official page says new projects are not being onboarded. Do not make launch depend on it or confuse Google login with health permission.
Health Connect / HealthKit [24, 25]	Optional device nutrition, activity and body metrics.	Plan native companion/bridge work. These are not ordinary browser OAuth data sources. Manual entry remains sufficient for the POC.
Garmin / Withings / other vendors	Device-specific metrics where a pilot cohort requests them.	Discovery candidates, not verified integrations. Compare approval, scopes, retention, overlap and ongoing cost first.
Calendar, contacts and reminders	Dinner attendance, prep reminders, shared household planning.	Start with user-controlled calendar file/export and in-app reminders. Add provider OAuth only for a demonstrated workflow.
Receipts, email and OCR	Unpack shopping with fewer scans.	Start with explicit user upload. Email connection needs narrow authorisation and retention rules; receipt lines rarely identify exact GTINs reliably.
REST / remote MCP [32]	Assistants and consumer tools can use current inventory, plans and dietary history.	Existing reads need live-client verification. Prioritise read freshness and per-household/per-profile scopes; add confirmed writes later.
Portable formats and household devices	CSV/JSON, Recipe JSON-LD, print, EPUB, private QR labels, Bluetooth/USB scanners and kitchen tablets.	Build on current exports. Add import contracts, versioning and keyboard-scanner support; no proprietary kitchen hardware required.

Give each connector a capability record: supported countries, read/write actions, data quality, source licence, cost, freshness, health permissions, access state and fallback. Show customers only actions available in their region; put provider diagnostics in the developer area.

Normalise once. Preserve the evidence.

Normalisation, hashing, compression, embedding and distribution do not remove source rights. Physical separation helps governance but does not itself settle derivative-database obligations.

Proposed data model

Entity	Essential fields and distinction
Product / product version	Internal ID; original GTIN string; canonical GTIN where valid; market; brand; pack size; label version; ingredient/allergen claims; source timestamp. A retail SKU is not a universal product ID.
Ingredient / composition record	Culinary identity, synonyms by locale, raw/cooked/drained form, edible portion, nutrient units/basis, portion or density evidence. Similar names do not prove nutritional equivalence.
Recipe / ingredient line	Recipe version; source text; parsed amount/range/unit; optionality; preparation; yield; mapped identity; parsing confidence; reviewed state; licensed instructions/images.
Lot / batch / serving / event	Household and location; received/opened/frozen dates; use-by/best-before/check-reminder separately; quantity basis; cook batch; served/eaten/saved/discarded events.
Offer / observation	Retailer and store; region; date; currency; pack quantity; price basis; loyalty condition; promotion; availability evidence; delivery/minimum-order conditions.
Source assertion / permission	Provider, source ID/URL, retrieval and effective times, licence/version, attribution, allowed uses/cache/export, transformation history, confidence and correction record.

Ingestion and publication pipeline

1. Register the source and approved uses. 2. Retain raw evidence only where permitted. 3. Parse and normalise into a source-labelled staging area. 4. Validate units, identities, duplicates and anomalies. 5. Send uncertain mappings to review. 6. Publish a versioned feature projection. 7. Refresh, withdraw and reprocess through the same auditable path.

Storage that fits the product

Keep licensed raw releases and large compressed snapshots in object storage where permitted. Use indexed operational tables for fast barcode, ingredient and household queries. Deliver small regional/offline packs and incremental updates. Retain commercial provider responses only within permitted bounds; do not dump whole third-party catalogues into every phone or user-state snapshot.

Choose evidence by field, not one universal winner

Use exact current package evidence for product-specific labels; reviewed regional composition for generic foods; explicit user corrections for that private lot. Retain conflicting assertions and show uncertainty. Version nutrient calculations so a later upstream correction does not silently rewrite dietary history.

Licence release gate: complete attribution, adapted-content obligations, permitted caching/export, commercial use, image rights and withdrawal handling. Never mix personal health records into a public share-alike data release. OFF, AFCD and commercial API obligations differ. [8-16]

Engineering for a household, not a snapshot

Retain the existing Cloudflare Worker/D1 application and Netlify identity path while proving the core flow. Avoid a platform migration merely to add features.

Recommended progression

Layer	Implementation direction
Client	Keep the responsive web/PWA surface. Split the large app module by domain; introduce typed contracts and migrations. Use IndexedDB for durable local operations and local snapshots; keep camera/receipt parsing responsive.
Identity and permissions	Separate User, Household, Membership and MemberProfile. An adult can edit shopping without seeing another adult's measurements. A child profile is not an autonomous account. Invitations expire; revocation applies across devices and integrations.
Operational data	Migrate the growing account JSON to household-scoped lots, recipe references, plans, events and grants. Keep revisioned snapshots for migration/backup, not as the only growing record.
Sync	Client mutation ID + server sequence + per-entity version. Queue offline, acknowledge durably, retry idempotently and resolve conflicts explicitly. Start with ordinary requests/polling; add realtime transport only where needed.
Inventory accounting	A cooking batch consumes raw ingredients once. Served portions draw from the batch. Eaten and plate waste partition a serving; unserved leftovers remain batch stock. Undo creates a compensating event.
Data enrichment	Server-side adapters, quotas, cache policy per source, background jobs and backoff. Separate public catalogue, private household and restricted-provider caches. Keep keys off the client.
API and MCP	One domain authorisation layer for UI, REST and MCP. Publish schemas, pagination, errors, rate limits and timestamps. Add safe proposed writes only after the same domain operations work in the UI.
Operations	Error monitoring without food/health payloads, provider health, import quality dashboard, restore drills, retention/deletion jobs, alerting and a rollbackable deployment. Measure cost per active household.

MCP compatibility is a living requirement

The reviewed current specification recommends Client ID Metadata Documents and retains dynamic client registration as a deprecated compatibility mechanism. OffMenu implements dynamic registration but not metadata-document registration. Assess the targeted clients, issuer handling and protected-resource discovery against the current specification; do not infer certification from unit tests. [32]

Practical performance targets

Proposed targets: cached scan acknowledgement within 300 ms; remote details progressively loaded; ordinary interactions responsive during scanning; mobile Core Web Vitals targets of LCP ≤2.5 s, INP ≤200 ms and CLS ≤0.1. Establish a measured baseline on midrange Android and iPhone before claiming compliance. Separate warm-cache and cold/offline cases.

Health features must earn trust

Design recommendations for general wellbeing. This review does not certify medical accuracy, food safety, privacy compliance or regulatory status.

Concern	Required product behaviour
Dietary restrictions	Store allergy, intolerance, preference, religious requirement and dislike separately. Evaluate product ingredients, traces/cross-contact and recipe substitutions. Missing evidence means “needs checking,” never “safe.”
Nutrition estimates	Preserve kcal/kJ and nutrient basis; sodium/salt conversions, raw/cooked state, serving yield and actual used product. Use generic fallback visibly. Percentage eaten scales a homogeneous dish estimate; mixed plates may require component-level correction.
Children and completion	Optional parent-controlled history and coarse completion choices. No deficit targets, weight-loss gamification, pressure to finish or peer comparison. Unknown age should not silently enable adult weight-loss recommendations.
Body and exercise information	Optional dated measurements with units/source, visibility controls, editable goals and separate permissions. Activity can inform convenience and scheduling; avoid automatically equating exercise calories with food allowance.
Dates and storage	Distinguish manufacturer use-by, best-before, opened-date instructions and an estimated check reminder. A barcode normally identifies a product; expiry needs label entry/OCR or a supported encoded date. [29, 33]
Privacy and consent	Document purposes; minimise collection; make sensitive sharing opt-in by person and destination. Allow export, deletion, connector revocation and local-data clearing on shared devices. Audit who accessed what without logging raw health content.
Wellness versus treatment	Describe food records, preferences and general wellbeing accurately. Disease-specific prescriptions, dysphagia decisions and diagnostic claims need separate clinical/regulatory assessment before promotion. [28]

Australian launch implications

OAIC says organisations providing a health service and holding health information may be covered by the Privacy Act even when small. OffMenu should assess its actual functions and claims rather than assume a startup exemption. Explain overseas providers and applicable handling of health information before enabling sensitive account flows. [27]

Separate safety from convenience

FSANZ distinguishes safety-based use-by dates from quality-based best-before dates. Do not generate a safe-to-eat deadline from “fruit lasts a week” or “drinks last two weeks.” Keep source-backed handling advice and user-selected review reminders separate. Batch recalls require the relevant batch/date evidence; otherwise show a potential match. [29, 30]

Suggested promotion language: “estimated nutrition,” “recorded intake,” “helps plan around your preferences,” and “check the pack.” Avoid “allergy-safe,” “clinically accurate,” “prevents disease,” or guaranteed health outcomes until those claims are substantiated.

A mature experience across people and places

Accessibility and internationalisation belong in the foundation. They should not depend on a later redesign or a premium subscription.

Capability	Design and verification requirement
Accessible interaction	Target WCAG 2.2 AA. Semantic forms, descriptive labels, keyboard order, visible/unobscured focus, skip links and usable dialogs. No essential drag-only operation or colour-only signal. [26]
Readable and adaptable	Text zoom and reflow, high contrast, large targets, reduced motion and sensible line lengths. Charts have a text/table equivalent. Use familiar language and clear error recovery.
Camera alternatives	Barcode typing, search, recent items and hardware scanner input. Show camera choice and permission recovery. Optional sound, speech, vibration and flash; no sensory channel is required.
Cognitive and motor needs	One task at a time, large primary actions, easy undo, saved drafts, quiet mode, low typing demand and no forced timed response. Test with disabled and neurodivergent users, not just automated checks.
Household realities	Attendance by meal, shift work, two homes, carers, guests, privacy on a shared tablet and travel. Multiple profiles do not automatically imply permission to expose health records.
Language and layout	Externalised strings, locale-aware plurals, accessible language metadata, longer translations, right-to-left layout and recipe language separate from interface language. Preserve original food names and accents.
Units and dates	Metric/imperial conversion that actually works; mass versus volume; regional spoon/cup conventions; count ranges; local date display, currency and time zone. Store unambiguous values with provenance.
Regional food behaviour	Country-specific aliases, cuisine, pack sizes, nutrient conventions, allergen rules and dietary guidance. Country choice is explicit. Do not assume a US composition record matches an Australian fortified product.
Network and device resilience	Offline inventory/list/recipes already downloaded, clear pending state, no silent data loss, retries, older-phone budgets and tablet/desktop layouts. Native wearable access stays optional.
Standard website expectations	Useful support and onboarding, privacy/terms/licences, account lifecycle, contact/report issue, status and recovery information, clear pricing/cancellation if paid, consent preferences and honest availability labels.

Minimum verification matrix

Android Chrome and iPhone Safari on physical phones; desktop keyboard; VoiceOver and TalkBack; narrow view/large text; denied camera; poor network; offline edits; shared-device logout; two adults editing the same list; revoked membership; an unknown barcode; conflicting food data. Record supported versions when testing rather than promise every device.

Use a locale pack to bind translated UI, supported units, food aliases, data sources, country guidance and retailer capabilities. Stage international expansion when that pack passes both usability and rights checks.

Promotion that demonstrates a result

Marketing recommendation: demonstrate a complete outcome, then measure whether households come back. Do not promote a directory of future connections.

Landing-page direction

Headline: Dinner starts with what you already have.
Support: Scan a few foods, find meals that fit your household, and share a shopping list for what is missing.
Primary action: Try a kitchen.
Secondary action: Add my first food.

Use a short visible demonstration: five foods, three options, one family meal, a shared list and tomorrow's leftovers. Add benefits, real screenshots, supported capabilities, source/trust details and FAQs below it. Keep detailed health and developer features on dedicated pages.

Channel / experiment	What to show	Success signal
Five-food challenge	A short, repeatable scan-to-dinner demo with actual supported foods.	Visitor reaches a useful meal and starts their own kitchen; count activated households, not video views alone.
Household invitation	Send a private list or invite someone to help plan, with clear shared scope.	Second person joins and contributes without support; no forced public sharing.
Food and family creators	Licensed recipe packs with practical adaptations, prep time and shopping gaps.	New households cook or plan from the pack and return the following week.
Search content	Useful, reviewed pantry meals and food-management guides with correct source rights and Recipe structured data where appropriate.	Organic visitors complete a relevant task. Do not index private kitchens or mass-publish incomplete AI pages.
Waste and budget pilots	Small local household cohort; compare their own recorded baseline and follow-up.	Less reported planning effort or measured discard/duplicate purchases. Describe self-report and uncertainty honestly.
Technical community	Documented read-only MCP demo, API sandbox, exports and a clear issue/reporting route.	A working authorised integration is reused. Keep this acquisition path separate from family onboarding.

Revenue hypotheses to validate

Keep basic scanning, accessibility, portability and the initial meal outcome easy to try. Test willingness to pay for sustained household convenience: reliable sync, richer planning, approved automation and history. Interview first, then test two clear price propositions without invented savings or fake checkout. Track retention and contribution margin before selecting a permanent price.

Optional retailer referrals and creator revenue share can fit, but disclose commissions and keep sponsorship out of safety filters and ranking logic. Do not fund the service by selling individual dietary or children's data. Avoid implying retailer, medical or Open Food Facts endorsement.

A staged plan with decision gates

Suggested sequencing for one experienced product engineer with access to content/design review. These are planning increments, not delivery guarantees; parallel content work can shorten elapsed time.

Increment	Concrete deliverable	Exit condition
1 - Repair the first impression First 2-3 working days	Curate the first ten everyday meals; repair hero demo; remove unresolved recipes from default recommendations; improve copy and empty states. Start measuring the main journey.	A new visitor reaches three credible options in the stocked demo without touching a data-review workflow.
2 - Prove useful data Following 3-5 working days	Reach 30 reviewed meals; about 100 ingredient mappings; stage regional composition fallback; add produce/manual food and ingredient corrections.	Required ingredients and yields quantified; all sources/rights recorded; representative products and generic foods mapped correctly.
3 - Make a household work Next 1-2 weeks, scope-dependent	Memberships, autosync, offline queue, cooking batches, leftovers, revocation and account lifecycle.	Two independent accounts on two physical devices complete scan, plan, shop, cook and after-meal updates with no lost or duplicate operations.
4 - Run an observed pilot Two-week usage window	Recruit 10-20 target households; use clear onboarding, feedback and optional event analytics. Benchmark Android/iPhone access and supported input methods.	Assess repeat meal outcomes, maintenance effort, support load and willingness to pay. Fix failure patterns before broad promotion.
5 - Expand on evidence	Add the most useful receipt/retailer flow, regional pack or health bridge; verify one real REST client and two representative MCP clients.	Demonstrated retention/coverage gain, approved source terms and measurable operating cost. Stop integrations that fail these tests.

Proposed pilot scorecard - decision targets, not forecasts

Measure	Initial target / decision use
Activation	At least 70% of observed target users complete add → useful meal → list/plan within five minutes, with no facilitator intervention.
Repeated value	At least 50% of activated pilot households complete two useful meal outcomes in week two. Use interviews to explain abandonment.
Data quality	100% of promoted recipes have reviewed required quantities and yields; every restriction unknown is visible; capture scan coverage and correction rate by retailer.
Reliability	No cross-household access, lost confirmed changes or duplicate stock deductions in the release scenarios. Record sync lag and camera failure rate by device.
Effort and economics	Measure minutes spent maintaining inventory per week, support contacts and provider/cloud cost per active household. Refine the product if bookkeeping outweighs benefit.

Start now: recipe/data quality, three-option dinner discovery, produce entry and trustworthy dietary status. In parallel as workstreams: household sync design and cooking/leftover accounting. A larger catalogue or more connection logos is not the next proof point.

Sources and review limits

Primary/provider sources accessed 12 October 2026. Bracketed numbers in the report refer to this register. Provider statements are not independent proof of product performance.

[1] Live OffMenu product Open source
Landing, demo menus and Shopping & apps inspected. No signed-in account or physical camera verification.

[2] Source archive / local evidence
offmenu-source.zip, Library version 4; SHA-256 6b931f25241640ba69aebd961a7316bae6871bf002de6b3d8e8ab362619f062e. Reviewed README, schema, app, models, handler and integrations. npm test: 41/41; Python unittest: 6/6. Catalogue counts and dietary counterexamples independently reproduced.

[3] Samsung Food+ Open source
Also reviewed Samsung Food meal-planner and FoodPlus promotional pages. Product-feature evidence, not independently verified outcomes.

[4] Paprika features and current beta Open source
Barcode capability is advertised in open beta; do not imply that every stable edition includes it.

[5] AnyList features Open source
Shared lists, meal planning and recipe management.

[6] Cronometer features Open source
Barcode and nutritional tracking comparison; no endorsement of accuracy claims.

[7] SuperCook Open source
Ingredient-based recipe discovery comparison.

[8] Open Food Facts licence guidance Open source
ODbL database and DbCL contents. Verify distinct image terms for any future image use.

[9] Open Prices API guidance Open source
API reuse and ODbL requirements.

[10] FSANZ AFCD data files Open source
Downloadable food composition records for software use.

[11] FSANZ database-specific licence Open source
Attribution, derivative work, limitations and Australian-data notices.

[12] USDA FoodData Central API guide Open source
Public-domain/CC0 data, key requirements, rate limits and downloads.

[13] Wikibooks cookbook dataset Open source
Dataset card declares CC BY-SA 4.0; individual provenance and image rights remain material.

[14] Edamam Recipe Search API Open source
Web recipe rights and restrictions on building/caching a copy of the service data.

[15] Spoonacular terms Open source
Reviewed terms dated 16 April 2026; transformed-storage, caching and competitive-use restrictions.

[16] FatSecret API editions Open source
Free editions show US-only datasets; international commercial access requires separate review.

[17] GS1 Australia NPC terms Open source
Also reviewed NPC/Trusted Content descriptions; consumer reuse is not assumed from catalogue access.

Sources and review limits

Primary/provider sources accessed 12 October 2026. Bracketed numbers in the report refer to this register. Provider statements are not independent proof of product performance.

[18] UK CoFID Open source
Regional composition candidate; check the exact downloaded release and licence.

[19] New Zealand FOODfiles terms Open source
Candidate only: search/FAQ evidence found, but full terms fetch timed out. No legal clearance asserted.

[20] ANSES Ciqual documentation Open source
Regional open-data candidate; exact release terms and required acknowledgement need implementation review.

[21] DoorDash Marketplace overview Open source
Limited-access partner/merchant APIs.

[22] Instacart Developer Platform Open source
Current documented shopping/recipe capabilities; key, market and flow verification still required.

[23] Google Health API Open source
Official current notice says new projects are not being onboarded.

[24] Android Health Connect Open source
Android SDK and app permissions; not a browser-native health data API.

[25] Apple HealthKit Open source
Native health framework; permission and supported-device implementation required.

[26] W3C WCAG 2.2 Open source
Accessibility target and normative success criteria; no conformance audit claimed.

[27] OAIC: health service providers Open source
Small-business status alone does not exclude health service providers from the Privacy Act.

[28] TGA general health/wellness software Open source
Actual intended purpose and features determine applicability; no regulatory determination made here.

[29] FSANZ food date labels Open source
Use-by safety versus best-before quality.

[30] FSANZ food recalls Open source
Authoritative recall notices; no complete machine-feed integration verified.

[31] FoodKeeper dataset record Open source
Storage-guidance candidate; verify current files, rights and regional applicability.

[32] MCP authorisation specification Open source
Protected-resource discovery, audience validation and evolving client registration requirements.

[33] GS1 2D barcodes Open source
Supported GS1 2D formats can carry identification and additional attributes such as expiry.

[34] WhichGrocer developer page Open source
Page returned a loading shell. Source code lists an access-required candidate; access, terms and exact capabilities remain unverified here.

Review limitations and retained decisions

This is a product/code review and a broad, prioritised opportunity assessment, not a penetration test, full accessibility audit, clinical review, legal opinion or market-size study. Source availability and terms can change. Commercial providers require their own agreements. No user accounts, provider credentials or health data were connected during this review.

The source archive was analysed without changing the product. The recommended roadmap is intended to make the current proof of concept credible, measurable and easier to use before further expansion.